Lecture– 10
Graphs Part 1

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Introduction to Graph

- In **Data Structures and Algorithms (DSA)**, a **graph** is a **non-linear data structure** that consists of:

Vertices (or nodes) — the entities.

Edges (or arcs) — the connections or relationships between the vertices

- A graph **G** is represented as:

G=(V,E) Where:

V is a set of vertices.

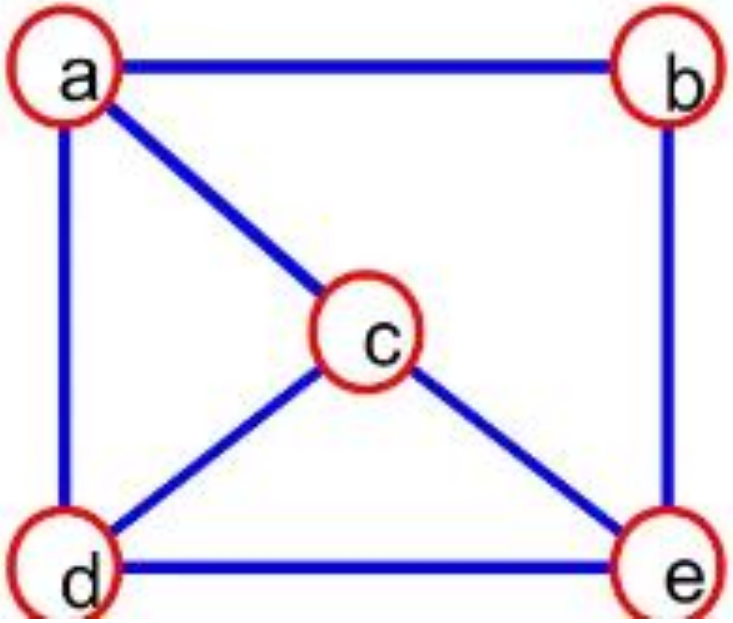
E is a set of edges, which are pairs of vertices.

What is a Graph?

- A graph G = (V,E) is composed of:

  V: set of vertices

  E: set of edges connecting the vertices in V
- An edge e = (u,v) is a pair of vertices
- Example:



V= {a,b,c,d,e}

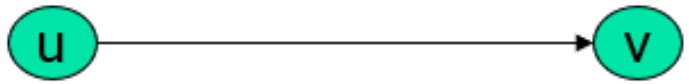
E= {(a,b),(a,c),
(a,d),
(b,e),(c,d),(c,e),
(d,e)}

Applications in Graphs

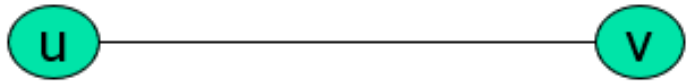
- **Computer Science Applications**
  - **Data Structures**: Representing networks such as social networks, web pages (links), and file systems.
  - **Routing Algorithms**: Dijkstra's, Bellman-Ford, and Floyd-Warshall for shortest paths in networks and more
- **Networking**
  - **Computer Networks**: Routers and switches are modeled as graph nodes; connections are edges.
  - **Internet Structure**: Web graph where web pages are vertices and hyperlinks are edges.
- **Social Networks**
  - Friendship/Follow Relationships: Facebook, Twitter, LinkedIn all use graph models.
  - Community Detection: Identifying clusters of people with common interests.
- **And more**

Different types of graphs

Directed: Ordered pair of vertices. Represented as (u, v) directed from vertex u to v.

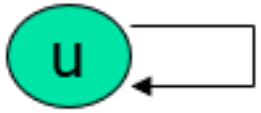


Undirected: Unordered pair of vertices. Represented as {u, v}. Disregards any sense of direction and treats both end vertices interchangeably.



Different types of graphs (cont..)

- **Loop:** A loop is an edge whose endpoints are equal i.e., an edge joining a vertex to it self is called a loop. Represented as {u, u} = {u}

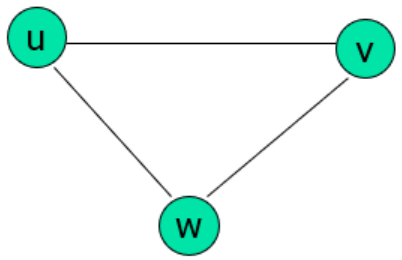


- **Multiple Edges:** Two or more edges joining the same pair of vertices.

Different types of graphs (cont..)

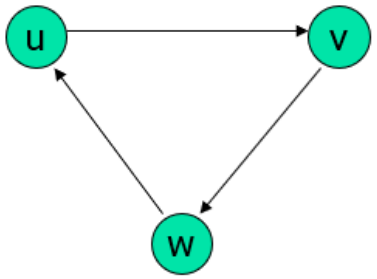
Simple (Undirected) Graph: consists of V, a nonempty set of vertices, and E, a set of unordered pairs of distinct elements of V called edges (undirected)

Representation Example: G(V, E), V = {u, v, w}, E = {{u, v}, {v, w}, {u, w}}



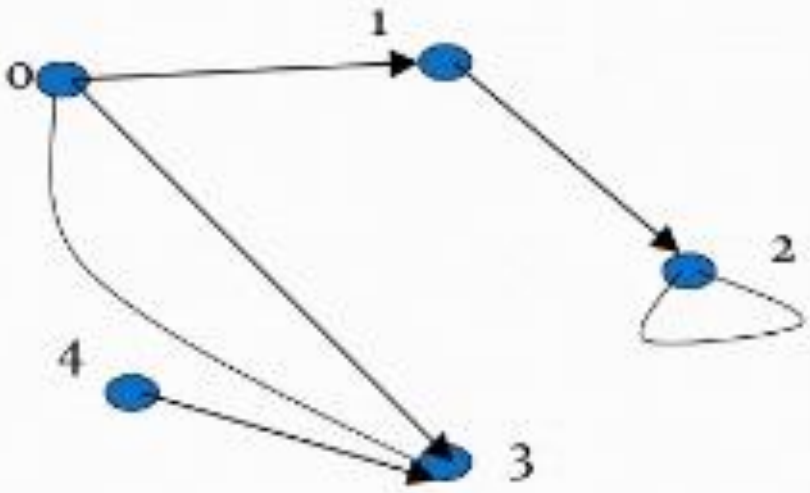
Directed Graph: G(V, E), set of vertices V, and set of Edges E, that are ordered pair of elements of V (directed edges)

Representation Example: G(V, E), V = {u, v, w}, E = {(u, v), (v, w), (w, u)}



Examples of Graphs

- V={0,1,2,3,4}
- E={(0,1), (1,2), (0,3), (3,0), (2,2), (4,3)}



When (x,y) is an edge,
we say that x is *adjacent to* y, and y is *adjacent from* x.

0 is adjacent to 1.
1 is not adjacent to 0.
2 is adjacent from 1.

Terminology – Adjacent and incident

- If (v_0, v_1) is an edge in an undirected graph,
  - v_0 and v_1 are adjacent
  - The edge (v_0, v_1) is incident on vertices v_0 and v_1
- If $<v_0, v_1>$ is an edge in a directed graph
  - v_0 is adjacent to v_1, and v_1 is adjacent from v_0
  - The edge $<v_0, v_1>$ is incident on v_0 and v_1

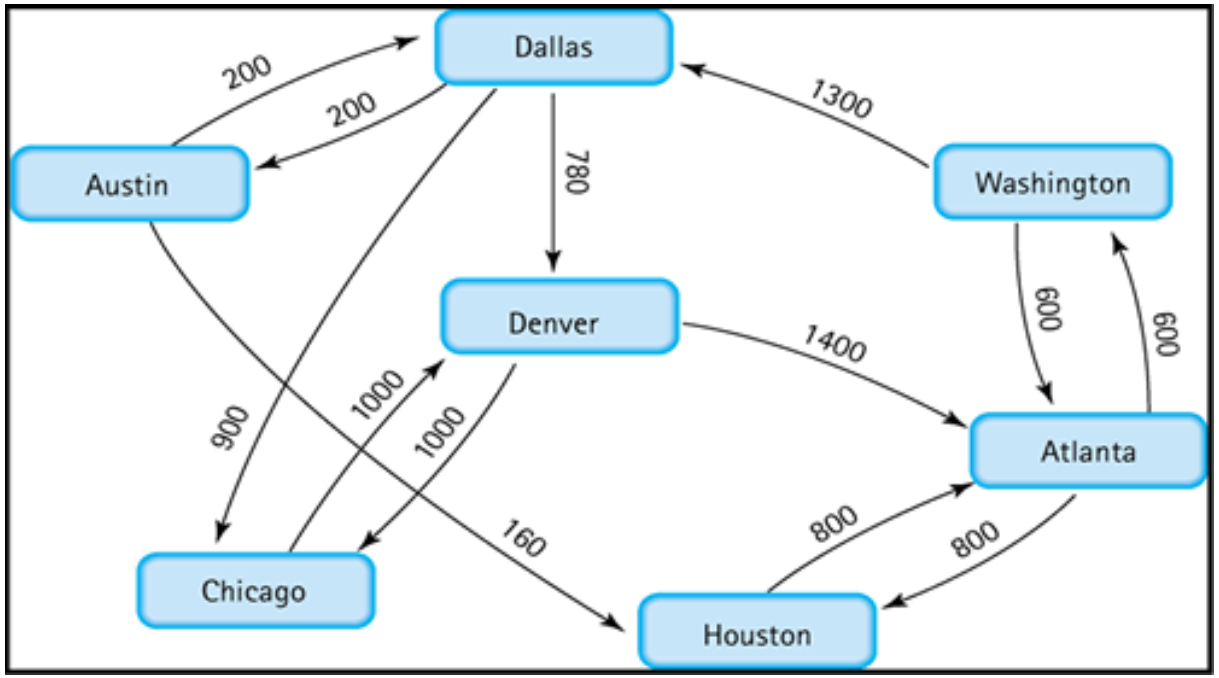
Graph –Example

- If numbers are associated with each arc of a graph, the graph is called a weighted graph or a network.

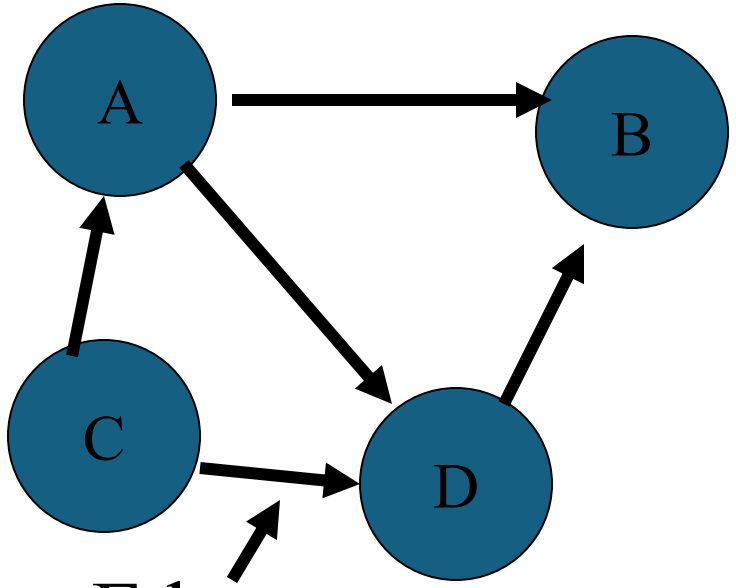
  Eg Nodes – towns

  Arcs - Road

  Weight – Distance



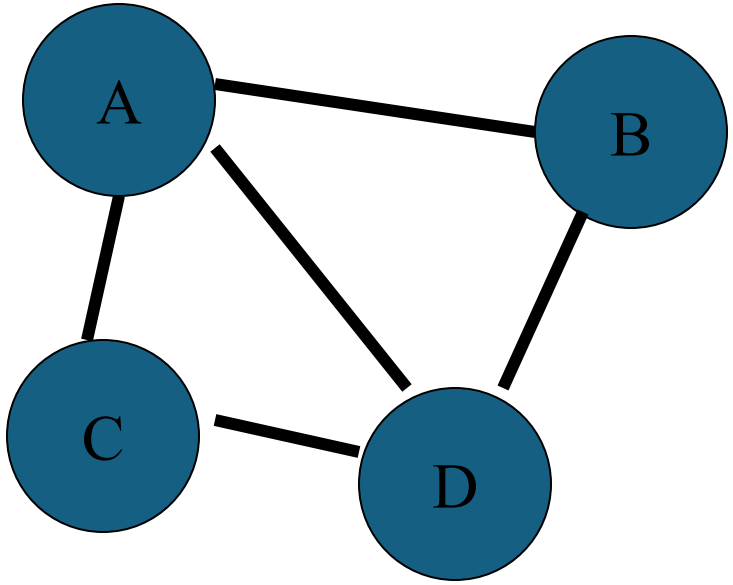
Examples



Nodes or vertex

Edges or arcs

Directed



Un-directed

Graph Terminology

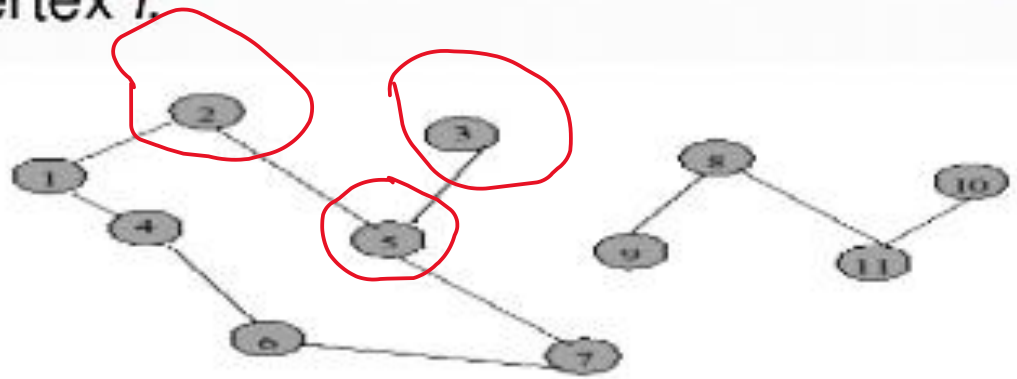
- Adjacent nodes: two nodes are adjacent if they are connected by an edge

5 is adjacent to 7
7 is adjacent from 5

- Path: a sequence of vertices that connect two nodes in a graph
- A simple path is a path in which all vertices, except possibly in the first and last, are different.
- Complete graph: a graph in which every vertex is directly connected to every other vertex

Graph Terminology cont..

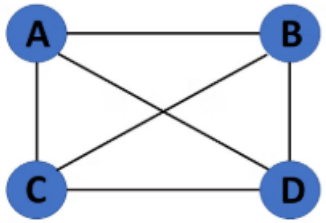
- A cycle is a simple path with the same start and end vertex.
- The **degree** of vertex *i* is the no. of edges incident on vertex *i*.



e.g., degree(2) = 2, degree(5) = 3, degree(3) = 1

Graph Terminology cont..

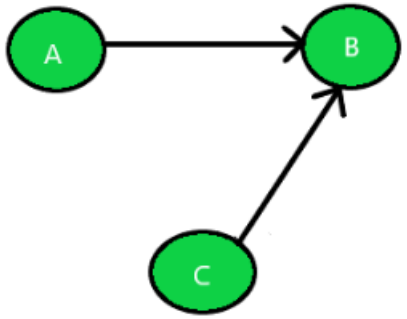
- Connected graph
  - A connected graph is a graph where every pair of vertices is joined by at least one path



- Strongly connected directed graph
  - A strongly connected directed graph is a directed graph where every vertex can reach every other vertex.

Graph Terminology cont..

- Weekly connected graphs : This property describes a directed graph where all vertices are connected if you disregard the direction of the edges



- **Complete digraph:** is a directed graph in which every pair of distinct vertices is connected by a pair of unique edges (one in each direction)

Example

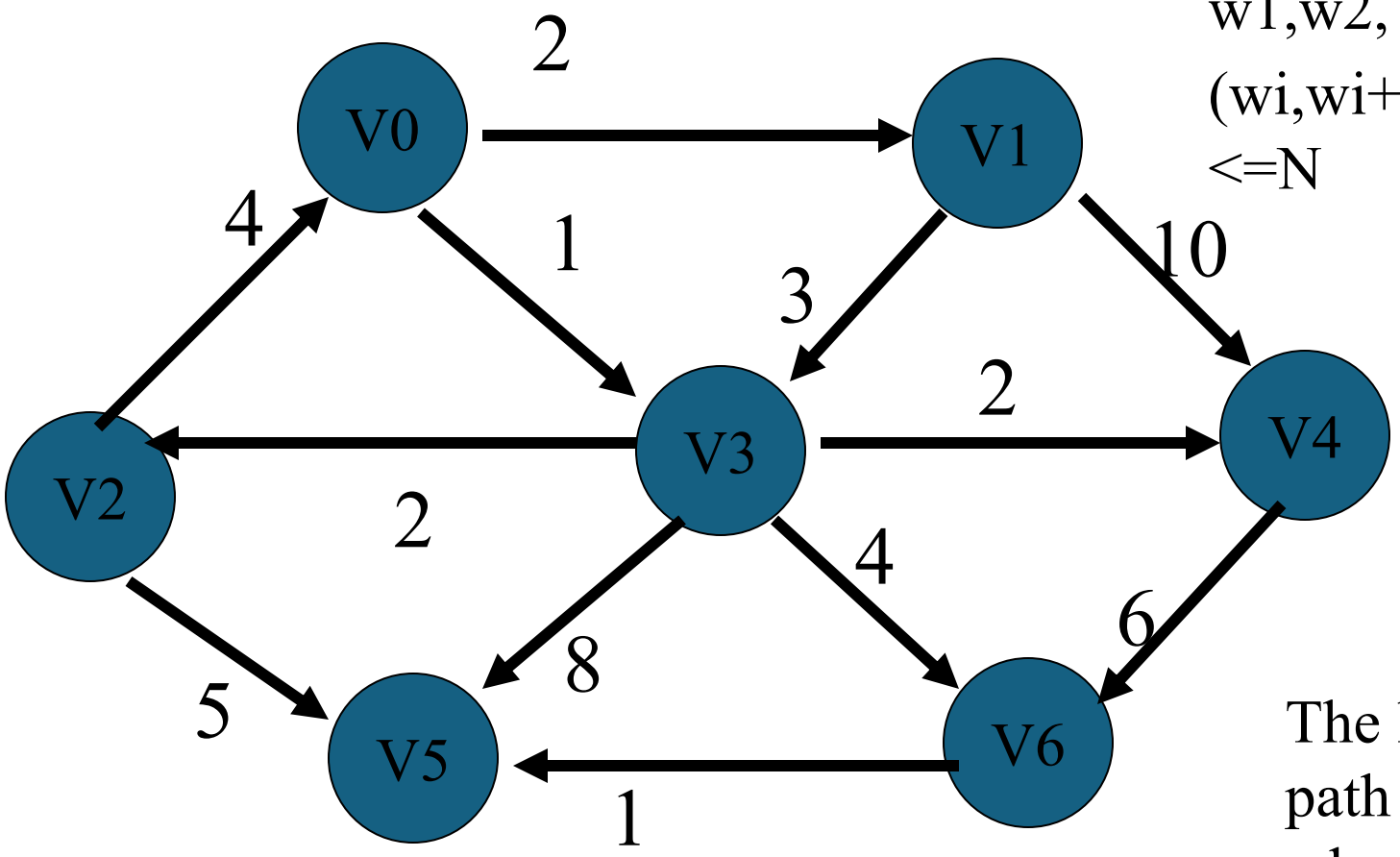
The graph G has the following 8 vertices.

V={
V0,V1,V2,V3,V4,V5,V6,V7 }

And 12 edges

E={(V0,V1,2),(V0,V3,1),(V1,V3,3),(V1.V4,10),V2,V4,2),(V3,V6,4),(V3,V5,8),(V3,V2,2),(V2,V0,4),(V2,V5,5),(V4,V6,6),(V6,V5,1)}

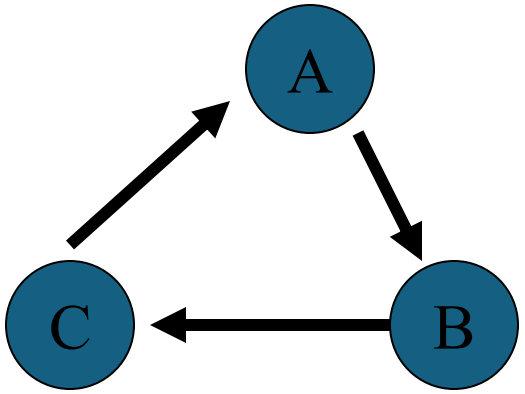
A path in a graph is a sequences of vertices w1,w2, …..wn such that (wi,wi+1) ε E for 1=< I <=N



A directed graph

The length of such a path is the number of edges on the path.(N-1). This is called un-weighted path length.

- The weighted path length is the sum of the costs of the edges on the path.
- As an example v0,v3,v5 is a path from vertex v0 to v5. The path length is two edges, and the weighted path length is 9.
- A path from a node to itself is called a cyclic. If a graph contains a cyclic, it is cyclic otherwise it is acyclic



Graph Implementations

- Graph data structures can be implemented in several ways depending on memory efficiency, type of operations, and use cases.
- The **main implementation methods** are:
- Adjacency Matrix
- Adjacency List
- Incidence Matrix
- Edge List/Table

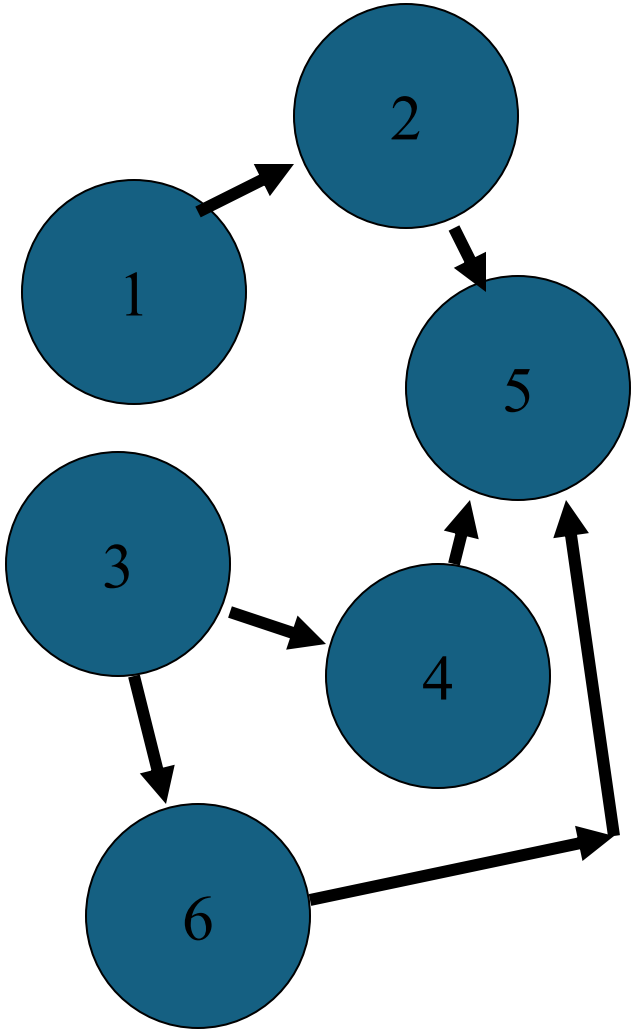
Adjacency Matrix in Graphs- (Array based implementation of graphs)

- An **adjacency matrix** is a **2D array (matrix) representation of a graph**.
- We can use a two-dimensional square array. The dimension of the array is equal to the maximum number of vertices that our graph can have.
- For a graph with V vertices, the adjacency matrix is a $V \times V$ matrix A where:
- For an **unweighted graph**:
  - A[i][j] = 1, if there is an edge between vertex i and vertex j
  - A[i][j] = 0, if there is no edge.
- For a **weighted graph**:
  - A[i][j] = w if there is an edge with weight w from i to j.
  - A[i][j] = 0 (or ∞ for directed graphs) if no edge exists

Adjacency Matrix in Graphs- (Array based implementation of graphs)

- In the simplest implementation of a graph, the array contains Boolean elements.
  In a graph represented by an array named *graph*, the element *graph[i][j]* is **true** if vertex *i* is connected to vertex *j* by an edge, and **false** if it is not.
  A purely array-based representation of a graph is called an **adjacency matrix**.
  The number of vertices in the graph is determined by the dimension of the array.

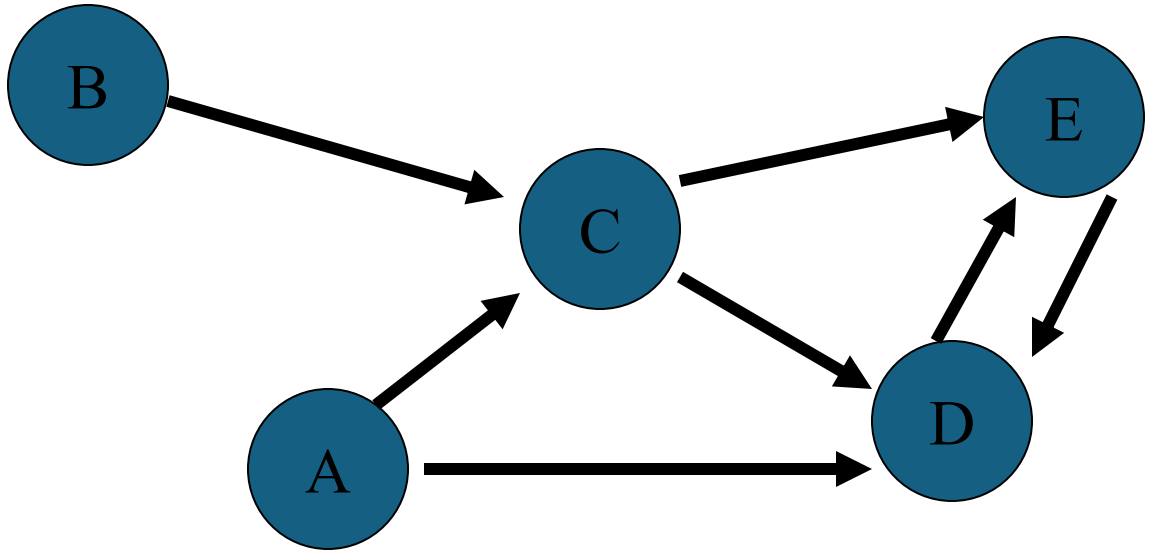
Consider the following unweighted directed graph



	1	**2**	**3**	**4**	**5**	**6**
1	F	T	F	F	F	F
2	F	F	F	F	T	F
3	F	F	F	T	F	T
4	F	F	F	F	T	F
5	F	F	F	F	F	F
6	F	F	F	F	T	F

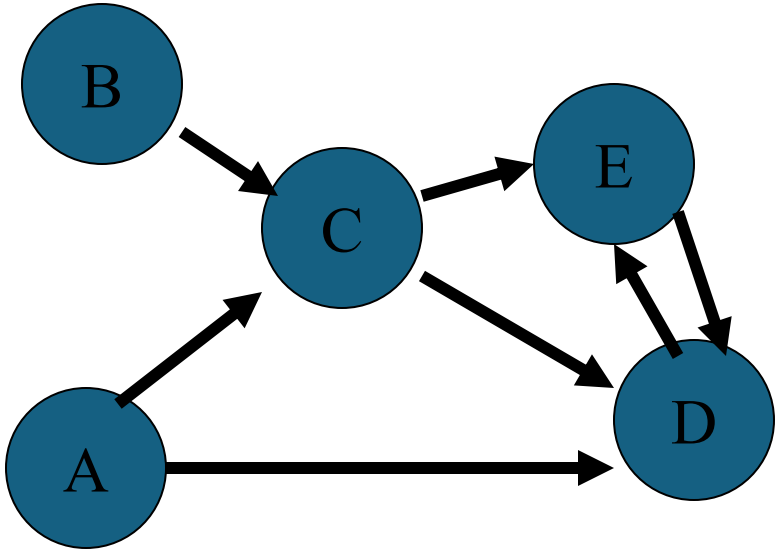
The graph is totally describing its adjacency matrix.

Boolean products of adjacency Matrices.



Adj[i][j] =1 or true then there
is an arc from node i to node j.
Adj[i][j] =0 or false otherwise

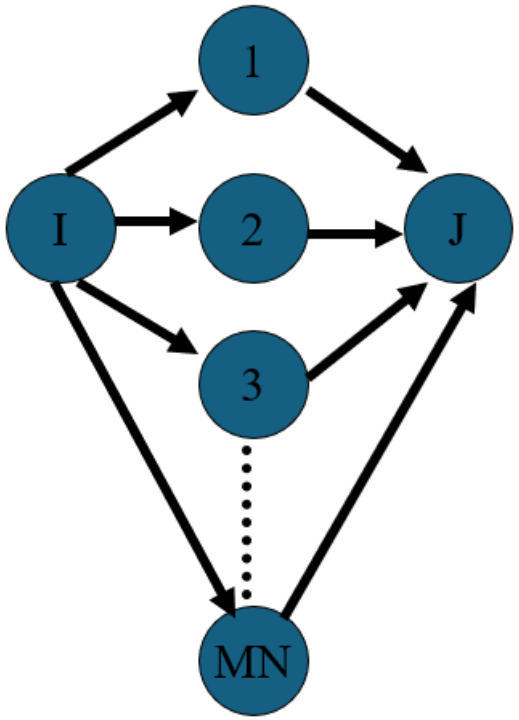
Let's us assume that a graph is completely describe its adjacency matrix



	A	**B**	**C**	**D**	**E**
A	0	0	1	1	0
B	0	0	1	0	0
C	0	0	0	1	1
D	0	0	0	0	1
E	0	0	0	1	0

ADj

Adjacency values



- Consider the logical expression
- Adj[i][k] and adj[k][j]. Its value is true if and only if the values of both adj[I][k] and adj[k][j] are true.which implies that there is an arc from node I to node j.
- Thus, adj[I][k] and adj[k][j] equal to if and only if there is a path of length 2 from node I to node j passing through k..
- If adj[I][j] is true if and only if (path of length=2)
- Adj[I][1] and adj[1][j] or
- Adj[I][2] and adj[2][j] or
- Adj[I][3] and adj[3][j] or

...

Adj[I][MN] and adj[MN][j]

The value of this expression is true only if there is a path of length 2 from node I to node j passing through node 1, node 2 ..... MN

Adjacency matrices

Adj_2 is said to be Boolean product of adj with self

Adj=path matrix with length = 1 [adj]

Adj_2=path matrix with length = 2 [adj*adj]

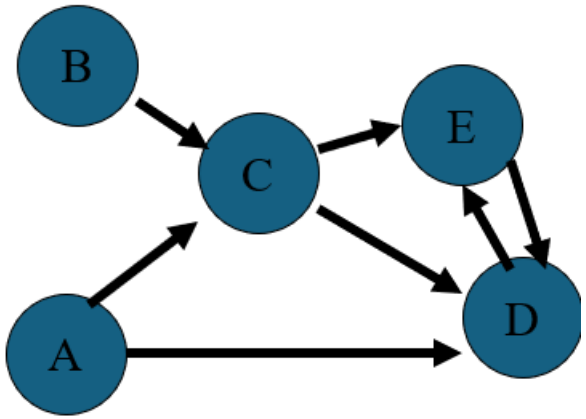
Adj_3=path matrix with length = 3 [adj_2*adj]

.....

Adj_l=path matrix with length = l [adj_{l-1}*adj

Question 1

- Consider the following directed graph.

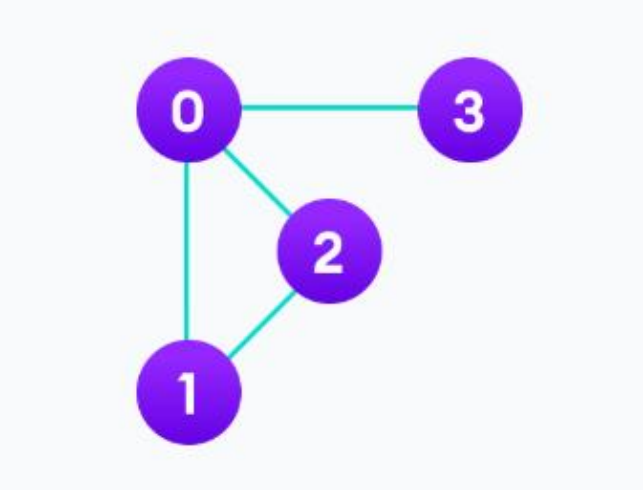


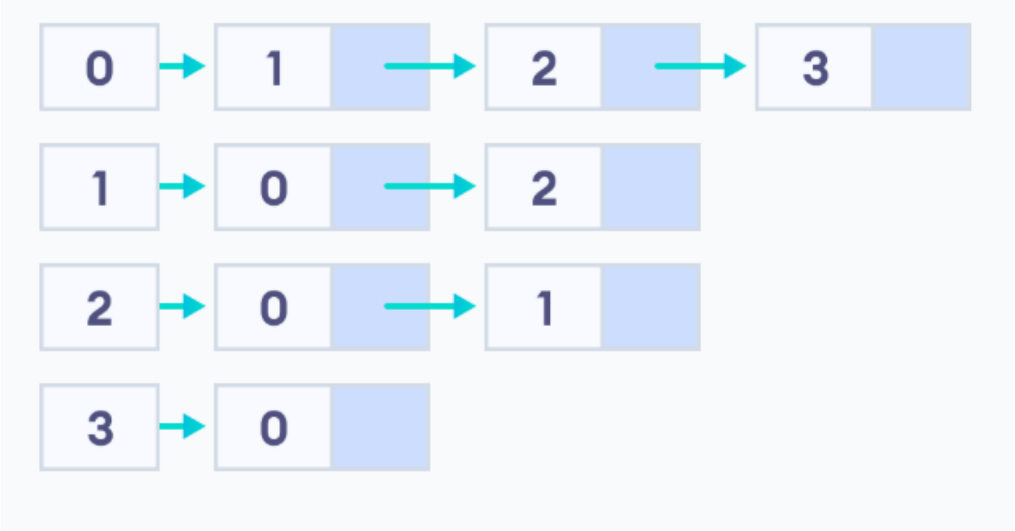
- Find the Adjacency matrices (adj, Adj_2,Adj_3,Adj_4 and Adj_5)
- Find the path matrix (transitive closure)

Advantages and disadvantages of adjacency matrix applications

Advantages	Disadvantages
Simple and Easy to Implement **(**Straightforward 2D array structure)	High Space Complexity (even if the graph has very few edges (sparse graph).
Constant-Time Edge Lookup	Inefficient for Sparse Graphs (Wastes memory when the number of edges E is much smaller than V^2)
Good for Dense Graphs (When the number of edges is close to the maximum possible (V^2), the matrix is efficient)	Slow to Traverse Neighbors
Supports Weighted Graphs (Can easily store edge weights instead of just 1/0)	Insertion/Deletion of Vertices is Costly (adding or removing a vertex requires resizing the matrix.)
Symmetry in Undirected Graphs (For undirected graphs, the matrix is symmetric, making it easier to process.)	Not Ideal for Certain Algorithms (Many graph algorithms (e.g., Dijkstra, BFS, DFS) are faster with adjacency)

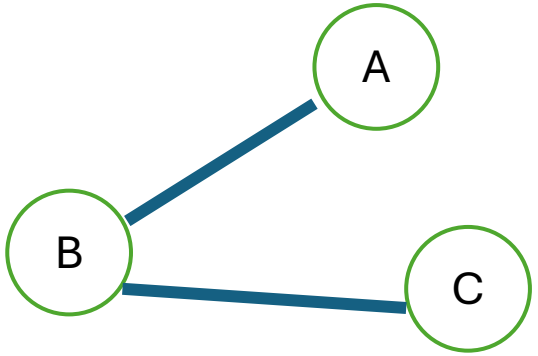
Adjacency lists representation of graphs

- An adjacency list represents a graph as an array of linked lists.
- The index of the array represents a vertex and each element in its linked list represents the other vertices that form an edge with the vertex.
- Example 1 – Undirected graph Adjacency List(Linked list of the graph)
- 



Incidence definition

- In **graph theory**, the word **incident** describes the relationship between a **vertex** and an **edge.**
  - An **edge** is said to be **incident on** the **vertices** that form its endpoints.
  - A **vertex** is said to be **incident with** an **edge** if that edge is connected to the vertex.
- Example:
- graph $G = (V, E)$ with
- V={A,B,C} and
- $E = \{(A, B), (B, C)\}$



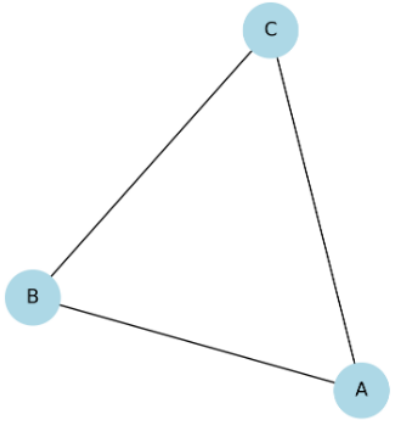
- The edge $(A, B)$is **incident** on vertices $A$and B.

Incidence Matrix (Graph Representation)

- An **incidence matrix** is a way to represent a graph using a **2D matrix** that shows the relationship between **vertices** and **edges**.
- Useful in **network flow problems, linear algebraic graph theory, and electrical circuits**.
- If a graph has V vertices and E Edges, its **incidence matrix** is a V×E matrix M , where:
  - Rows represent **vertices**.
  - Columns represent **edges**.
  - M[i][j]= 1(or sometimes −1) if vertex i is **incident** to edge j (i.e., edge j is connected to vertex i).
  - Otherwise, M[i][j]=0

Incidence Matrix (Graph Representation)

- Example
- Consider the following undirected graph.

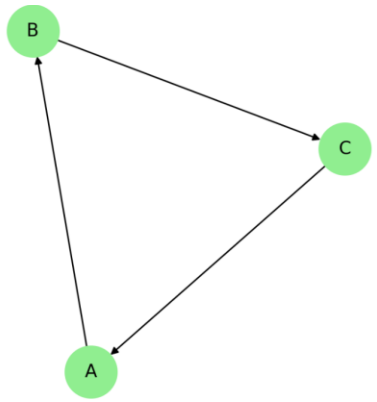


- Vertices: {A, B, C}
- edge: {e1 = A–B, e2 = B–C, e3 = C–A}

Vertex/Edge	e1 (A–B)	e2 (B–C)	e3 (C–A)
A	1	0	1
B	1	1	0
C	0	1	1

Incidence Matrix (Graph Representation)

- Example
- Consider the following directed graph.

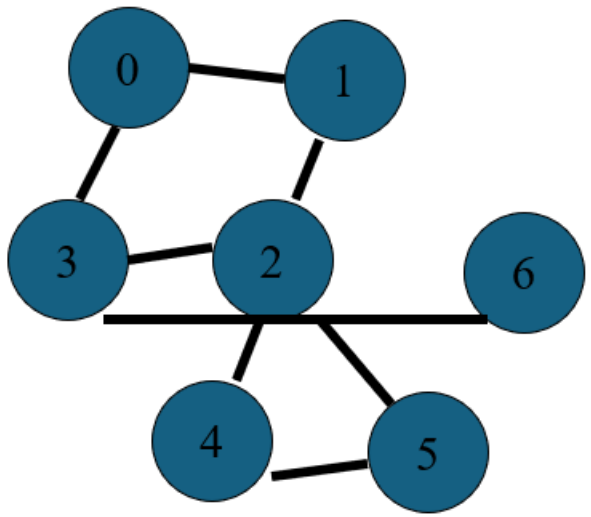


- Vertices: {A, B, C}
- Edges: {e1 = A → B, e2 = B → C, e3 = C → A}
- Incidence Matrix:

Vertex/Edge	e1 (A→B)	e2 (B→C)	e3 (C→A)
A	-1	0	1
B	1	-1	0
C	0	1	-1

Edge list/Table

Consider the following undirected graph



Node	Vertex list
0	1 3
1	0 2
2	1 3 4 5
3	0 2 6
4	2 5
5	2 4
6	3

Graph Traversal

- The most basic graph algorithm that visits nodes of a graph in certain order.
- There are two commonly used methods of travelling a graph.
  - Depth First Traversal (DFT): use recursion stack
  - Breadth First Traversal (BFT): use queue

Depth-First Traversal/ Search (DFS/DFT)

DFS(v): visits all the nodes reachable from v in depth-first order

- Mark v as visited
- For each edge $v \to u$:
  - If u is not visited, call DFS(u)

- Use non-recursive version if recursion depth is too big (over a few thousands)
  - Replace recursive calls with a stack

Breadth-First Traversal/ Search (DFS/DFT)

BFS(v): visits all the nodes reachable from v in breadth-first order

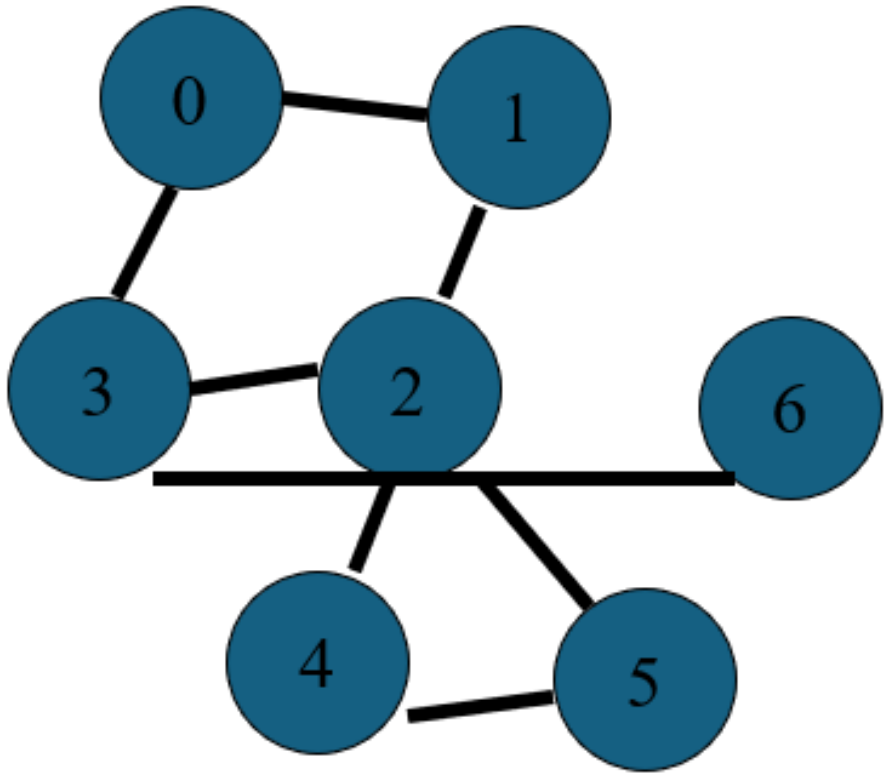
- Initialize a queue Q
- Mark v as visited and push it to Q
- While Q is not empty:
  - Take the front element of Q and call it w
  - For each edge $w \to u$:
    - If u is not visited, mark it as visited and push it to Q

Depth First Traversal (DFT)

- First, we construct an edge table for the graph.
- An edge table is built by listing all the vertices of the graph in a vertical column.
- Next to each entry in the column we list all vertices to which that entry is connected directly.

Question 2

- Consider the following Graph.
- Creating the edge table and traversing it using Depth First Traversal.

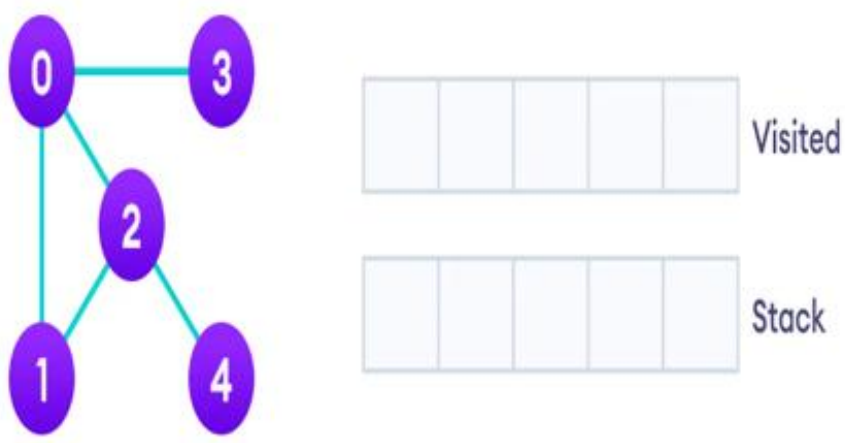


Implementation of Depth First Search/Traversal using a Stack

- **Depth First Search Algorithm**
- A standard DFS implementation puts each vertex of the graph into one of two categories:
  - Visited
  - Not Visited
- The purpose of the algorithm is to mark each vertex as visited while avoiding cycles.
- The DFS algorithm works as follows:
  - Start by putting any one of the graph's vertices on top of a stack.
  - Take the top item of the stack and add it to the visited list.
  - Create a list of that vertex's adjacent nodes. Add the ones which aren't in the visited list to the top of the stack.
  - Keep repeating steps 2 and 3 until the stack is empty.

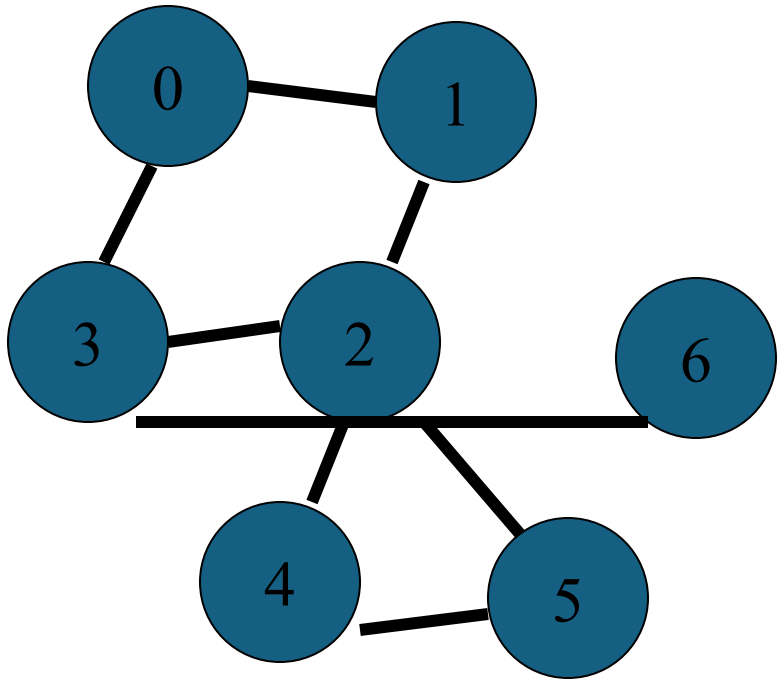
Question 3

- Consider the following Graph.



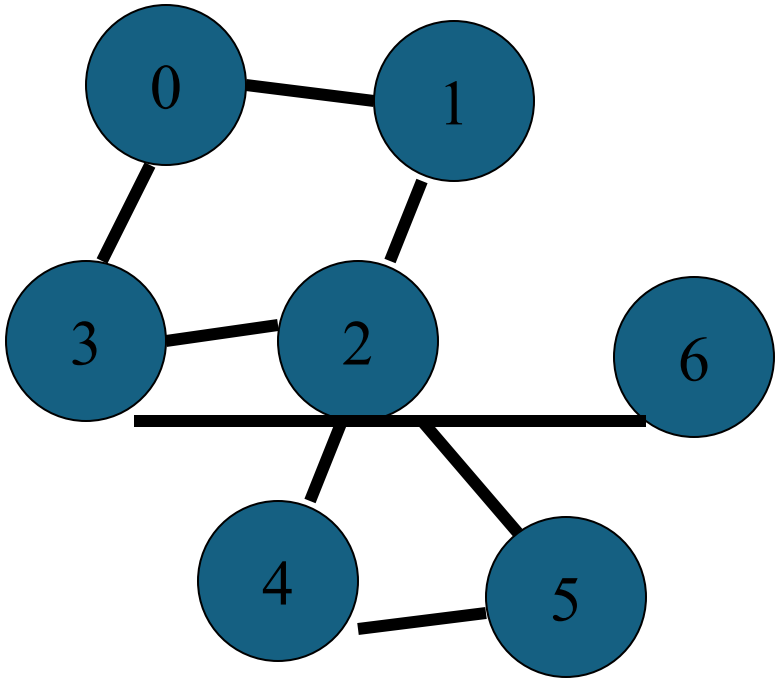
- Explain how to traverse the above graph using Depth First Traversal with a stack.
- Write the pseudocode algorithm for DFT/DFS

Breadth First Traversal



The idea here is to list all the vertices connected to each vertex before moving the on in the edge table.

Example (BFS/T)



The edge table for the above graph.

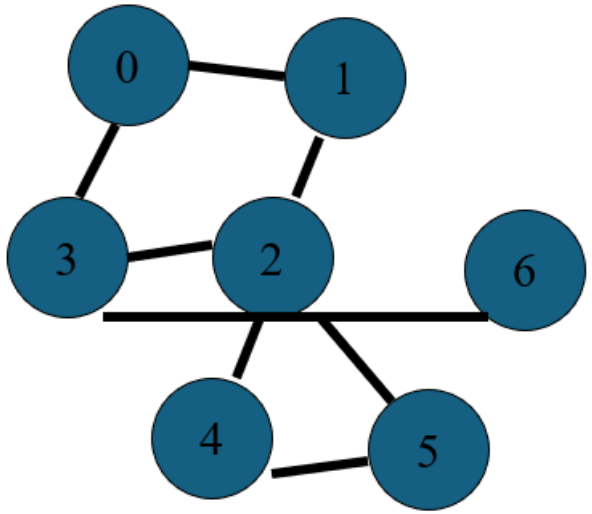
Node	Vertex list
0	1 3
1	0 2
2	1 3 4 5
3	0 2 6
4	2 5
5	2 4
6	3

Algorithm

1. Start at vertex V and mark it visited
2. Visit all the un-visited vertices adjacent to V
3. Visit all the un-visited adjacent to the vertices visited at the previous level
4. Continue this process until all the vertices are visited.

Question 4

- Consider the following graph.



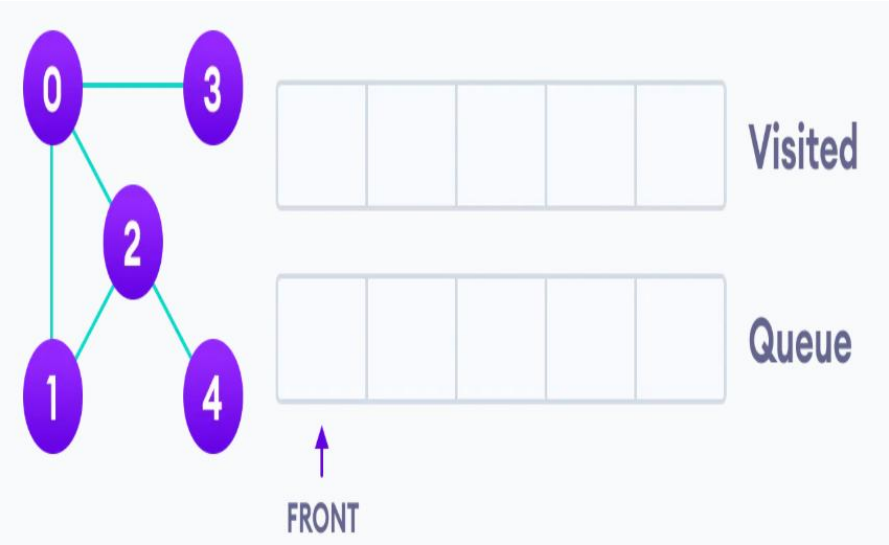
- Creating the edge table and traversing it using Breadth First Traversal.

Implementation of **BFS** Search/Traversal using a Queue

- The algorithm works as follows:
- **Step 1**: Start by putting any one of the graph's vertices at the back of a queue.
- **Step 2**: Take the front item of the queue and add it to the visited list.
- **Step 3**:Create a list of that vertex's adjacent nodes. Add the ones which aren't in the visited list to the back of the queue.
- **Step 4**: Keep repeating steps 2 and 3 until the queue is empty.

Question 5

- Consider the following Graph.



- Explain how to traverse the above graph using Breadth First Traversal with a queue.
- Write the pseudocode algorithm for BFT/BFS

Lecture – 11
Graphs –Part 2

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Spanning Tree

- A **Minimum Spanning Tree (MST)** is a fundamental concept in graph theory.
- A **spanning tree** of a connected, undirected graph is a subgraph that:
  - Connects all the vertices of the graph.
  - Has no cycles.
  - Contains exactly **n – 1 edges** (where *n* = number of vertices)
- Minimum Spanning Tree (MST)

  •Among all possible spanning trees of a weighted graph, the **MST** is the one with the **minimum possible total edge weight**.

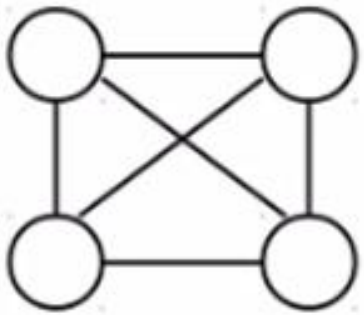
  •If a graph has multiple spanning trees, there may be more than one MST if the edge weights allow.

Spanning Tree

- Suppose you have a connected, undirected graph.
  - **Connected**: every node is reachable from every other nodes.
  - **Undirected**: edges do not have an associated direction.
- A **spanning tree** of a graph is a connected subgraph in which there are no cycles.
- A spanning tree of a graph is cycle-free but still connects all the vertices.
- If G is a connected graph with n vertices and m edges, spanning tree of G must have n-1 edges, and no of edges deleted from G to get a spanning tree must be m-(n-1) = m-n+1

Question 1

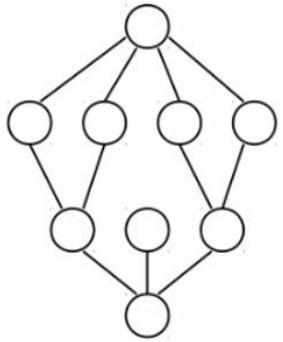
- Find all the spanning trees of the following graph.



A connected, undirected graph

Question 2

- Consider the following undirected graph.



An undirected graph

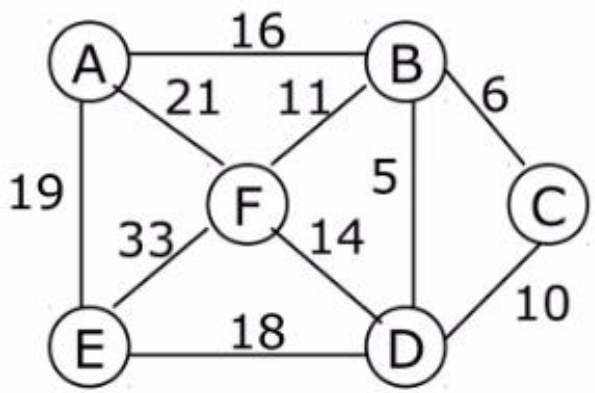
- Construct one spanning tree of the graph based on BFS and DFS traversal

Minimizing the Cost

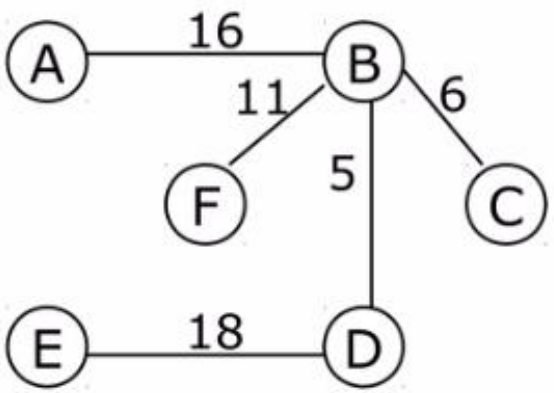
- Suppose you want to supply a set of houses (say, in a new subdivision) with:
  - electric power
  - water
  - sewage lines
  - telephone lines
- To keep costs down, you could connect these houses with a spanning tree (of, for example, power lines)
  - However, the houses are not all equal distances apart
- To reduce costs even further, you could connect the houses with a *minimum-cost* spanning tree

Minimum Cost spanning trees

- Suppose you have a connected undirected graph with a weight (or cost) associated with each edge
- The cost or weight of a spanning tree would be the sum of the costs of its edges.
- A minimum-cost spanning tree is a spanning tree of a connected undirected graph that has the lowest cost.
- If there are n vertices in the graph , then each spanning tree has n vertices & n-1 edges.



A connected, undirected graph



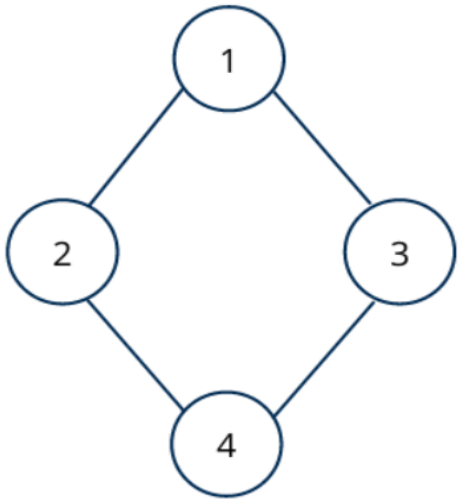
A minimum-cost spanning tree

Application of Minimum Spanning Tree

- Cluster Analysis
- Handwriting recognition
- Image segmentation
- Civil Network Planning
- Network Routing

Question 3

- Consider the following undirected connected graph:

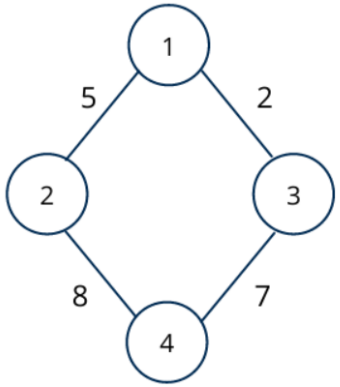


- Draw all possible spanning trees

Question 4

a) Define what is the minimum spanning tree?

- Consider the following graph:



b) List all possible spanning trees

c) Find the minimum spanning tree

Finding minimum spanning trees

There are two basic algorithms for finding minimum-cost spanning trees, and both are greedy algorithms i.e., these always process the edge with the least weight or cost.

- Kruskal's algorithm: Start with no nodes or edges in the spanning tree and repeatedly add the cheapest edge that does not create a cycle.
- Here, we consider the spanning tree to consist of edges only.

- Prim's algorithm: Start with any one node in the spanning tree, and repeatedly add the cheapest edge, and the node it leads to, for which the node is not already in the spanning tree.
- Here, we consider the spanning tree to consist of both nodes and edges

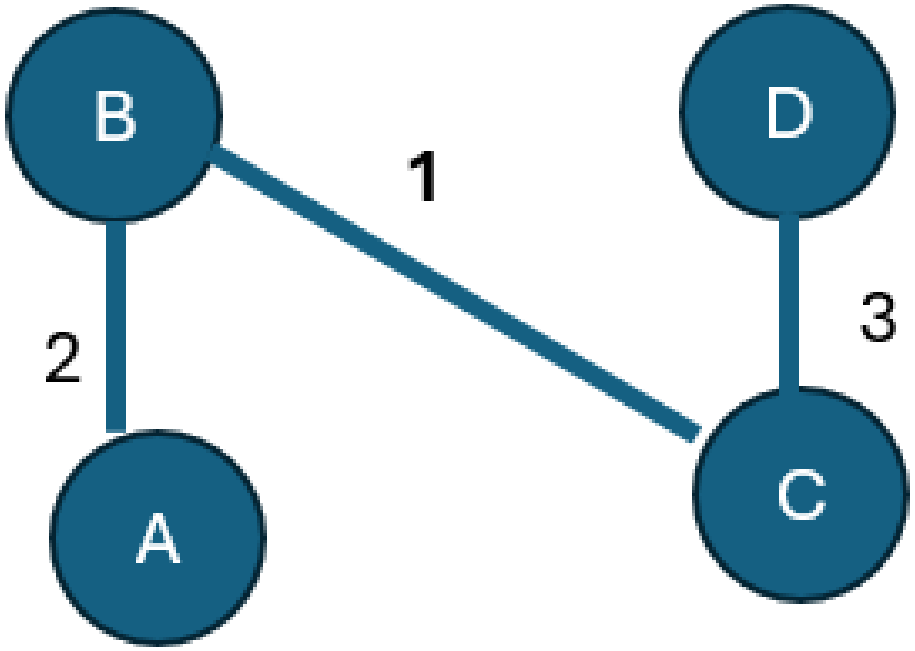
Kruskal’s algorithm

- Joseph Kruskal was an American Mathematician, Statistician and Computer Scientist.
- He was born on January 29, 1928
- In Computer Science, his best-known work is Kruskal's Algorithm for computing the Minimal Spanning Tree (MST) of a weighted graph.

Kruskal's algorithm

- Kruskal's algorithm is an algorithm for computing MST of a weighted graph.
- A MST is spanning tree of connected, undirected graph which connect all the vertices together with the minimum total cost weightage of its edges and it always have a V-1 edges.

- Example



Pseudocode - Kruskal’s algorithm

- The steps for finding MST using Kruskal's algorithm

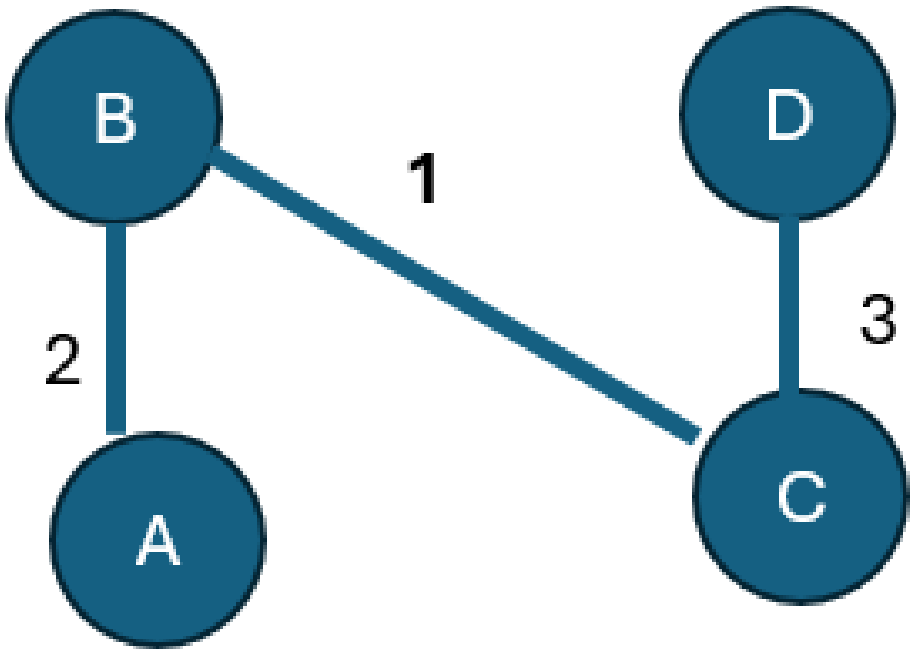
Step 1 : Sort all the edges in a non-decreasing order of their weight.

Step 2: Pick the smallest edge. Check if it forms a cycle with the spanning tree formed so far. If the cycle is not formed, include this edge. Else, discard it.

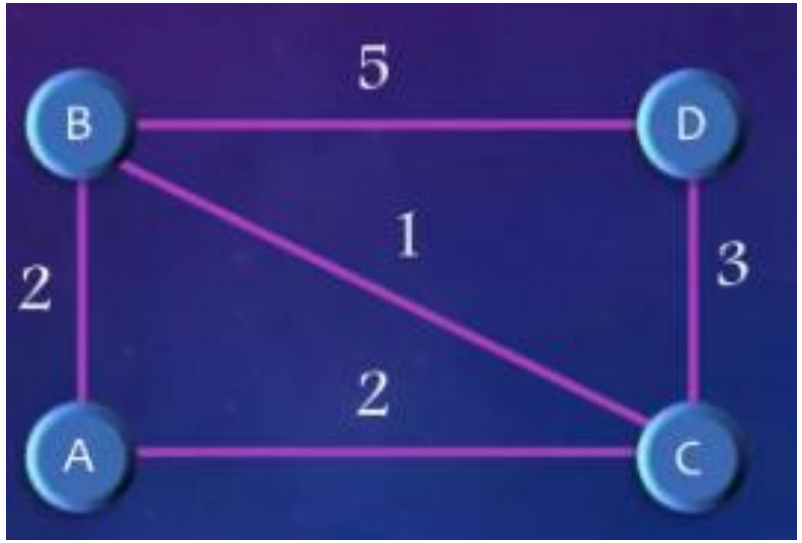
Step 3: Repeat step 2 until there are (V-1) edges in the spanning tree.

How to find minimal spanning tree (MST) using Kruskal's algorithm?

- we will always start from the smallest weight edge and keep selecting edges that does not from any circuit with the previously selected edges.



Steps to find MST by using Kruskal's Algorithm



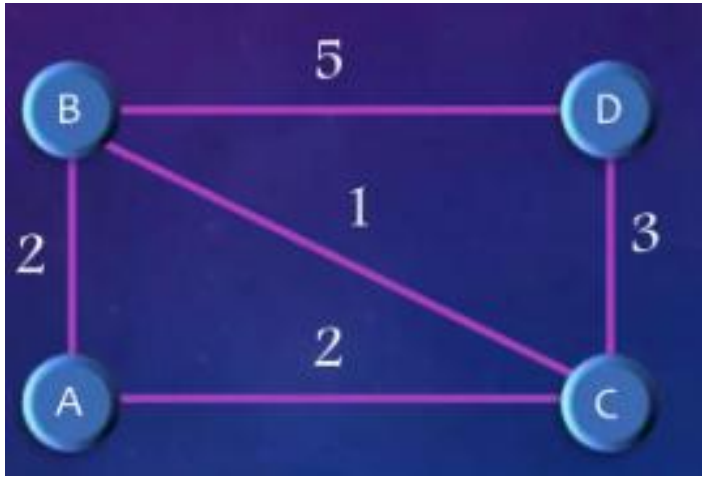
- We will always start from the smallest weight edge and keep selecting edges that does not form any circuit with previously selected edges.
- Edge table

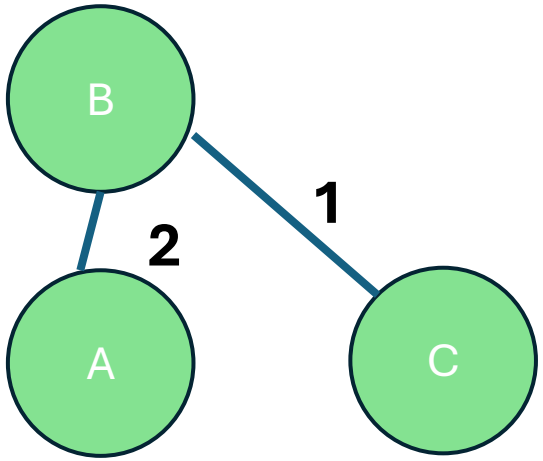
Edge					
Weight					

- We have five edges. So, our edge table will have 5 columns.
- Create the edge table (an edge table will have name of all the edges along with weight in ascending order)

Steps to find MST by using Kruskal's Algorithm

- Fill the first column with the edge of minimum weight
- B-C is the minimum weight (weight=1)
- Next minimum weights are B-A (weight=2)
- **Pick next smallest edge:** A-C (2)
- X skip, it forms a cycle with A–B–C



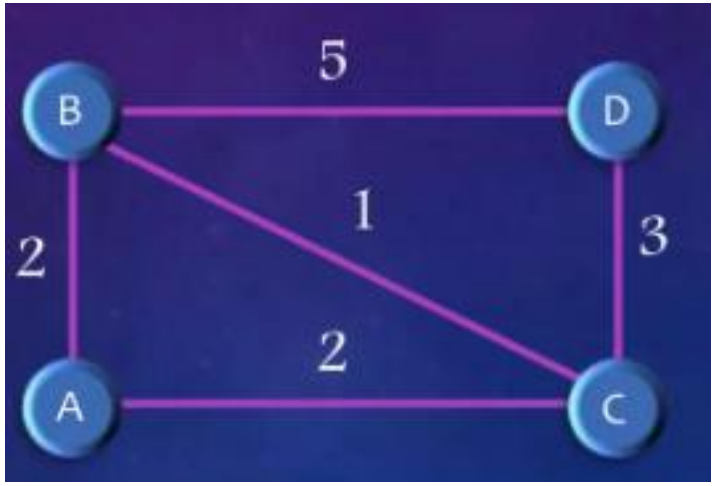


Edge	B-C	B-A				
Weight	1	2				

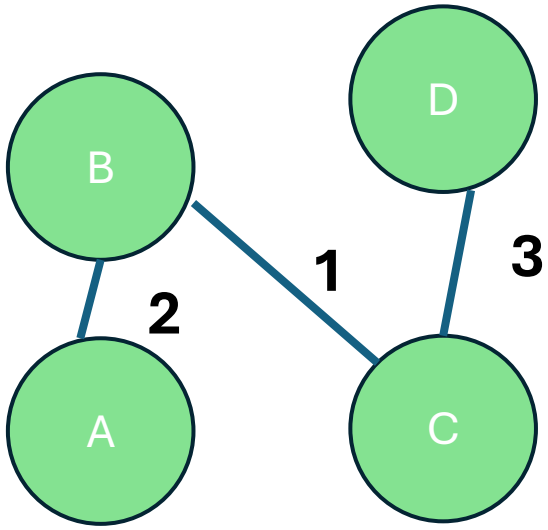
Steps to find MST by using Kruskal's Algorithm

- **Pick next smallest edge:** C-D (Weight=3)

Edge	B-C	B-A	C-D			
Weight	1	2	**3**			

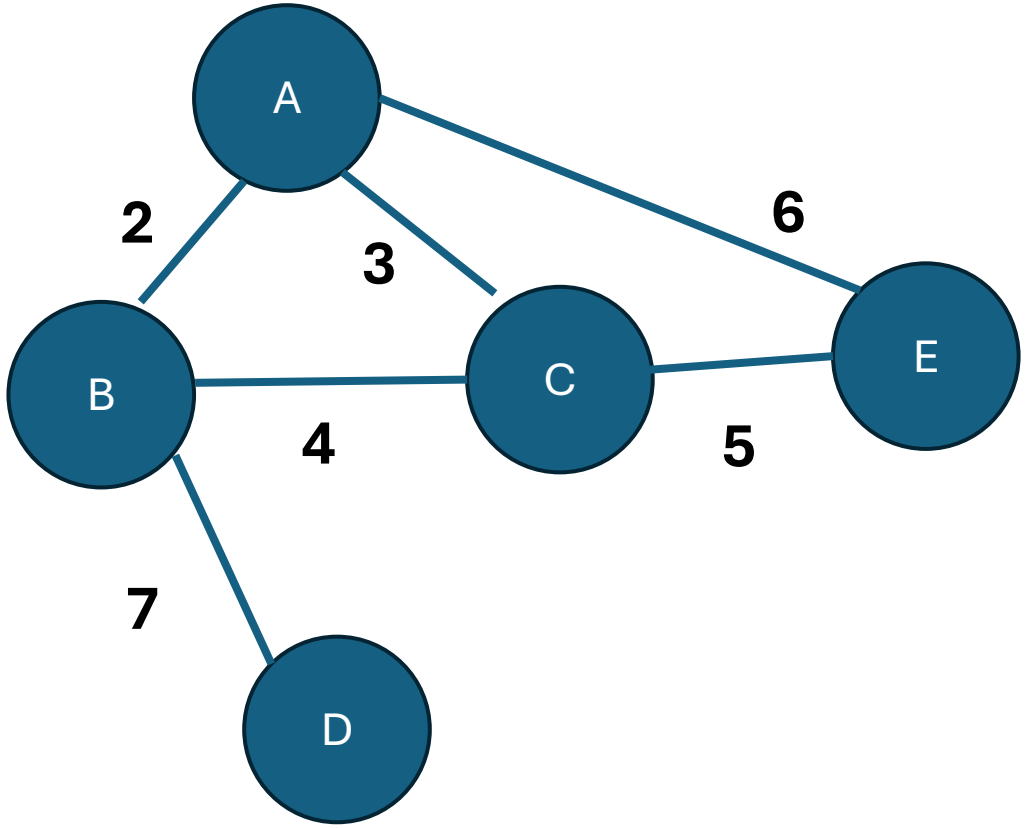


Total Minimum Weight = 1+2+3=6



Question 5

- How to find minimal spanning tree (MST) using Kruskal's algorithm?



Prim’s Algorithm for Minimum Spanning Tree (MST)

- Prim’s Algorithm is a **greedy algorithm** used to find the **Minimum Spanning Tree (MST)** of a connected, undirected, weighted graph.
- A **spanning tree** is a subgraph that connects all vertices of the graph without forming cycles, and the **minimum spanning tree** is the spanning tree with the minimum possible total edge weight.
- Deference between the **Prim’s Algorithm and** Kruskal’s Algorithm
- **Kruskal’s Algorithm**: *Sort all edges by weight and keep adding the cheapest edge that doesn’t form a cycle*
- **Prim’s Algorithm**: *Start from one node and grow the MST step by step by adding the cheapest edge leading out.*

Steps of Prim's Algorithm

Step 1 :Initialize

- Declare an array visited[] (or a set) to keep track of visited vertices.
- Select an arbitrary root vertex, say S, and mark it as visited.
- Initialize the MST result as empty.

Step 2 : Find Minimum Edge

- From all edges that connect **a visited vertex** to **an unvisited vertex**, choose the edge with the **minimum cost**.

Step 3 : Update MST

- Add this least-cost edge to the MST.
- Mark the newly connected vertex as visited.

Step 4 : Repeat

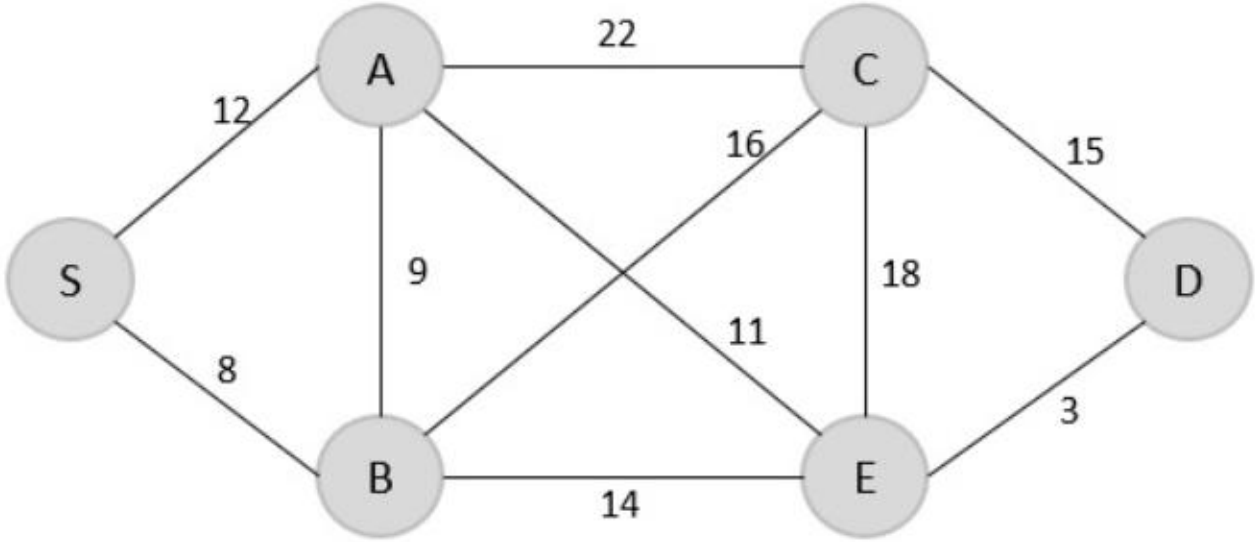
- Repeat steps 2 and 3 until all vertices are visited (i.e., MST contains $V - 1$edges for a graph with Vvertices).

Step 5 : Calculate Cost

- Sum the weights of the edges in the MST to get the total minimum cost

Question 6

- Consider the following graph



1. Find the minimum spanning tree using prims method (greedy approach) for the graph given below with S as the arbitrary root.
2. Write a C program to find the Minimum Spanning Tree (MST) using Prim's Algorithm